Does the Inertia of a Body Depend upon its Energy-Content?

by A. Einstein

September 27, 1905

The results of a recent electrodynamical investigation lead to a very interesting conclusion, which shall here be deduced. That investigation rested on the Maxwell–Hertz equations for empty space, together with the Maxwellian expression for the electromagnetic energy of space, and in addition the principle of relativity.

Let a system of plane waves of light, referred to the coordinate system *(x, y, z)*, possess the energy l; let the direction of the ray make an angle φ with the axis of x. If we introduce a new system of coordinates *(ξ, η, ζ)* moving in uniform parallel translation with respect to the first, and having its origin in motion along the axis of x with velocity v, then this quantity of light — measured in the new system — possesses the energy

$$l^* = l\,\frac{1-(v/c)\cos\varphi}{\sqrt{1-v^2/c^2}},$$

where c denotes the velocity of light. Let there now be a stationary body in the system, and let its energy — referred to *(x, y, z)* — be E_0, while its energy referred to the moving system *(ξ, η, ζ)* is H_0. Let this body emit, in a direction making an angle φ with the axis, plane waves of light of energy ½L, and simultaneously an equal quantity in the opposite direction, so that it remains at rest.

Denoting by K_0 and K_1 the kinetic energies of the body before and after emission, and neglecting magnitudes of fourth and higher orders, the difference between the two is given, to the required approximation, by the strikingly simple relation

$$K_0 - K_1 = \tfrac{1}{2}\,\frac{L}{c^2}\,v^2.$$

The kinetic energy of the body with respect to the moving system diminishes, as a result of the emission, by an amount that is independent of the properties of the body itself and depends only upon the energy given off. Since the difference $K_0 - K_1$ has, moreover, the same form as the kinetic energy of a mass point, we may write

$$K_0 - K_1 = \tfrac{1}{2}\Delta m\, v^2, \quad \text{so that} \quad \Delta m = L/c^2.$$

If a body gives off the energy L in the form of radiation, its mass diminishes by L/c^2. That the energy withdrawn becomes energy of radiation evidently makes no difference, so that we are led to the more general conclusion:

The mass of a body is a measure of its energy-content; if the energy changes by L, the mass changes in the same sense by $L/9\times10^{20}$, the energy being measured in ergs and the mass in grammes.

It is not impossible that with bodies whose energy-content is variable to a high degree — radium salts, for instance — the theory may be successfully put to the test. If it corresponds with the facts, radiation conveys inertia between the emitting and the absorbing bodies alike.

∗ The principle of the constancy of the velocity of light is, of course, contained in Maxwell's equations. Translated from "Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig?", *Annalen der Physik* **18** (1905), pp. 639–641.